

# VALUATION-GCD DYNAMICS ON PRIME-POWER DIVISORS: EXACT ENTRY TIMES AND EVERY-TARGET FIBRES

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**ABSTRACT.** Fix an odd prime  $p$  and  $e \geq 2$ . On the divisor chain of  $p^e$ , consider  $d \mapsto \gcd(p^e, d^2 + p^e/d)$ . Under  $p^a \leftrightarrow a$ , the literal integer map is exactly  $a \mapsto \min(2a, e - a)$ . We give its complete finite atlas. The recurrent exponents are 0 together with the integer band from  $\lceil e/3 \rceil$  to  $\lfloor 2e/3 \rfloor$ ; the band consists of fixed points and complement two-cycles. We prove a four-case pointwise entry-time formula, an exact temporal polynomial, and a sharp maximum  $1 + \lceil \log_2 \lceil e/3 \rceil \rceil$ . For  $e \geq 4$ , the unique deepest divisor is  $p^{e-1}$ ; the cases  $e = 2, 3$  have the unique deepest divisor  $p^e$ . The image is an interval in valuation coordinates, and every target fibre is the set union of at most two explicit inverse branches. Oddness is necessary: when  $p = 2$  and  $e = 3a$ , equal valuations create one extra factor of two. General valuation algebra, functional-graph bookkeeping, interval-map theory, and discretized tent maps receive zero contribution credit. Exact computation is used only for falsification, and external release remains on hold.

## 1. SCOPE AND COMPLETE STATEMENT

Piecewise-monotone interval maps have an extensive general theory [3], and finite tent-type discretizations have been studied in both reversible and bijective forms [1, 2]. Those theories receive zero contribution credit here. This note concerns one literal arithmetic self-map and does not claim that its piecewise-linear silhouette is new.

Fix an odd prime  $p$  and an integer  $e \geq 2$ . Define

$$(1) \quad \mathcal{X}_{p,e} = \{p^a : 0 \leq a \leq e\}, \quad F_{p,e}(d) = \gcd\left(p^e, d^2 + \frac{p^e}{d}\right).$$

Put

$$(2) \quad L = \left\lceil \frac{e}{3} \right\rceil, \quad U = \left\lfloor \frac{2e}{3} \right\rfloor, \quad R = U - L + 2, \quad A = 1 + \mathbf{1}_{2|e}.$$

For a positive rational number  $x$ ,  $\lceil \log_2 x \rceil$  is used below only when  $x \geq 1$  and means the least integer  $j \geq 0$  with  $2^j \geq x$ .

Under the identification  $p^a \leftrightarrow a$ , let  $T_e$  denote the exponent map. The entry time  $\tau_e(a)$  is the least  $t \geq 0$  for which  $T_e^t(a)$  is recurrent. The following is the claim ceiling for the note.

**Theorem 1.1** (Complete arithmetic temporal and inverse atlas). *For every odd prime  $p$  and  $e \geq 2$ , the following hold.*

(i) *The literal valuation identity is*

$$(3) \quad F_{p,e}(p^a) = p^{T_e(a)}, \quad T_e(a) = \min(2a, e - a).$$

(ii) *The recurrent exponent set is*

$$(4) \quad \text{Rec}(T_e) = \{0\} \cup \{L, L+1, \dots, U\}.$$

*It has  $R$  elements, of which  $A$  are fixed; all other recurrent exponents form strict complement two-cycles. For every  $k \geq 1$ ,*

$$(5) \quad |\text{Fix}(T_e^k)| = \begin{cases} A, & k \text{ odd}, \\ R, & k \text{ even}. \end{cases}$$

(iii) *The complete pointwise entry law is*

$$(6) \quad \tau_e(a) = \begin{cases} 0, & a = 0 \text{ or } L \leq a \leq U, \\ \lceil \log_2(L/a) \rceil, & 1 \leq a < L, \\ 1, & a = e, \\ 1 + \lceil \log_2(L/(e-a)) \rceil, & U < a < e. \end{cases}$$

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Consequently

$$(7) \quad M_e := \max_{0 \leq a \leq e} \tau_e(a) = 1 + \lceil \log_2 L \rceil.$$

For  $e \geq 4$ , the unique deepest exponent is  $e - 1$ ; for  $e = 2, 3$ , it is  $e$ .

(iv) Let  $m = \lceil \log_2 L \rceil$  and, for  $1 \leq j \leq m$ , put

$$(8) \quad c_j = \left\lfloor \frac{L}{2^{j-1}} \right\rfloor - \left\lfloor \frac{L}{2^j} \right\rfloor.$$

Then the complete temporal polynomial is

$$(9) \quad D_e(z) := \sum_{a=0}^e z^{\tau_e(a)} = R + z + (1+z) \sum_{j=1}^m c_j z^j.$$

(v) The image and every-target fibres are

$$(10) \quad \text{im } T_e = \{0, 1, \dots, U\},$$

and, for  $0 \leq b \leq e$ ,

$$(11) \quad T_e^{-1}(b) = \begin{cases} \{e-b\} \cup (\{b/2\} \text{ if } 2 \mid b), & 0 \leq b \leq U, \\ \emptyset, & U < b \leq e. \end{cases}$$

The union is a set: its two displayed candidates coincide exactly when  $3b = 2e$ .

The next four sections prove the five parts of [theorem 1.1](#). No finite enumeration is used in those proofs.

## 2. LITERAL VALUATION AND THE BINARY BOUNDARY

**Lemma 2.1** (Odd-prime valuation identity). *For  $0 \leq a \leq e$ , equation (3) holds.*

*Proof.* The argument inside the gcd is

$$(12) \quad (p^a)^2 + \frac{p^e}{p^a} = p^{2a} + p^{e-a}.$$

Set  $q = \min(2a, e - a)$ . If  $3a \neq e$ , the two exponents in (12) differ, and factoring out the smaller one gives

$$p^{2a} + p^{e-a} = p^q (1 + p^{|3a-e|}).$$

The factor in parentheses is not divisible by  $p$ . If  $3a = e$ , then

$$p^{2a} + p^{e-a} = 2p^{2a} = 2p^q,$$

and 2 is again a  $p$ -adic unit because  $p$  is odd. Thus the valuation of (12) is always  $q$ . Since  $0 \leq q \leq e$ , taking the gcd with  $p^e$  gives  $p^q$ , which is (3).  $\square$

*Remark 2.2* (Sharp characteristic-two obstruction). Oddness cannot be removed from [lemma 2.1](#). If  $p = 2$  and  $e = 3a$  with  $a \geq 1$ , then

$$2^{2a} + 2^{e-a} = 2^{2a} + 2^{2a} = 2^{2a+1}.$$

Because  $2a + 1 \leq 3a = e$ , the literal gcd has valuation  $2a + 1$ , rather than  $\min(2a, e - a) = 2a$ . The smallest witness is

$$\gcd(2^3, 2^2 + 2^2) = 2^3.$$

This note uses the binary calculation only as a boundary; it claims no replacement atlas for  $p = 2$ .

## 3. THE RECURRENT BAND AND FIXED ITERATES

The elementary identities

$$(13) \quad e - U = L, \quad e - L = U$$

follow directly from (2). They make the two branches of  $T_e$  especially rigid.

**Lemma 3.1** (Branch and first-entry geometry). *The exponent map satisfies*

$$(14) \quad T_e(a) = \begin{cases} 2a, & 0 \leq a < L, \\ e - a, & L \leq a \leq e. \end{cases}$$

*The interval  $[L, U]$  is invariant and is acted on by  $a \mapsto e - a$ . Moreover, if  $1 \leq a < L$  and  $j$  is least with  $2^j a \geq L$ , then  $2^j a \leq U$ .*

*Proof.* The two arguments in the minimum (3) satisfy  $2a \leq e - a$  exactly when  $3a \leq e$ . Every integer  $a < L = \lceil e/3 \rceil$  has  $3a < e$ , while every  $a \geq L$  has  $3a \geq e$ , proving (14). If  $L \leq a \leq U$ , then (13) gives  $L \leq e - a \leq U$ , so the band is invariant and the stated complement rule holds there.

It remains to exclude an overshoot by the last doubling. Minimality of  $j$  gives  $2^{j-1}a < L \leq 2^j a$ , so  $2^j a < 2L$ . If  $e = 3q$ , then  $(L, U) = (q, 2q)$ ; if  $e = 3q + 2$ , then  $(L, U) = (q + 1, 2q + 1)$ . In either case the integer  $2^j a < 2L$  is at most  $U$ . If  $e = 3q + 1$ , then  $(L, U) = (q + 1, 2q) = (L, 2L - 2)$ . Here  $j \geq 1$ , so  $2^j a$  is even. It is less than  $2L$  and therefore cannot equal the odd integer  $2L - 1$ ; hence it is at most  $2L - 2 = U$ .  $\square$

**Theorem 3.2** (Complete recurrence and fixed iterates). *Equations (4) and (5) hold.*

*Proof.* Zero is fixed. By lemma 3.1, every positive exponent below  $L$  doubles until its first entry into  $[L, U]$ . If  $a > U$ , its first image is  $e - a < L$ ; this image is zero only when  $a = e$ , and otherwise the preceding doubling argument applies. Thus every exponent enters the set displayed in (4).

On  $[L, U]$ , the map is the involution  $a \mapsto e - a$ . Hence every point of that band is recurrent. Conversely, every state outside the displayed set has strictly positive entry time, so it cannot be recurrent. This proves (4). Its size is  $1 + (U - L + 1) = R$ .

Within the band, a fixed exponent satisfies  $2a = e$ , so it exists exactly when  $e$  is even and is then  $e/2$ . Together with zero, this gives  $A$  fixed states. Every other band exponent is in a strict pair  $\{a, e - a\}$ . A transient point cannot be fixed by a positive iterate. An odd iterate fixes only the  $A$  fixed states, whereas an even iterate fixes all  $R$  recurrent states, proving (5).  $\square$

For completeness, define the formal Artin–Mazur zeta function by  $\zeta_e(z) = \exp(\sum_{k \geq 1} |\text{Fix}(\mathbf{T}_e^k)| z^k / k)$ . The cycle census in theorem 3.2 immediately gives

$$(15) \quad \zeta_e(z) = (1 - z)^{-A} (1 - z^2)^{-(R-A)/2}.$$

Equation (15) is standard finite-map bookkeeping and receives zero contribution credit.

#### 4. POINTWISE TIMES AND THE TEMPORAL POLYNOMIAL

**Theorem 4.1** (Pointwise entry time and sharp uniqueness). *Equations (6) and (7) hold, including the stated unique deepest exponents.*

*Proof.* The entry time is zero on the recurrent set. If  $1 \leq a < L$ , then lemma 3.1 shows that the orbit is  $a, 2a, 4a, \dots$  until its first band entry. The least number of doublings needed is the least  $j$  such that  $2^j a \geq L$ , namely  $\lceil \log_2(L/a) \rceil$ . The last assertion of lemma 3.1 ensures that this first crossing lies in the recurrent band.

If  $U < a < e$ , the first step sends  $a$  to the positive lower exponent  $b = e - a < L$ , after which the lower formula applies. If  $a = e$ , the first image is zero. These cases prove (6).

Put  $m = \lceil \log_2 L \rceil$ . Lower exponents have depth at most  $m$ . Upper exponents other than  $e$  have the form  $e - b$  with  $1 \leq b < L$  and depth  $1 + \lceil \log_2(L/b) \rceil$ , so their depth is at most  $1 + m$ . The choice  $b = 1$ , or  $a = e - 1$ , attains  $1 + m$ . Suppose  $e \geq 4$ ; then  $L \geq 2$  and  $m \geq 1$ . For  $b \geq 2$ ,

$$\frac{L}{b} \leq \frac{L}{2} \leq 2^{m-1},$$

so its upper partner has depth at most  $m$ . The lower branch also has depth at most  $m$ , and  $a = e$  has depth one. Thus  $e - 1$  is the unique exponent of depth  $1 + m$ , proving (7) and uniqueness. When  $e = 2$  or  $3$ , one has  $L = 1$ ; the recurrent set contains every exponent except  $e$ , which maps to zero in one step. Hence  $e$  is the unique deepest exponent in both small cases.  $\square$

**Theorem 4.2** (Complete temporal polynomial). *Equation (9) holds.*

*Proof.* For  $1 \leq j \leq m$ , a lower exponent  $a$  has depth exactly  $j$  if and only if

$$2^{j-1}a < L \leq 2^j a.$$

Equivalently,

$$\left\lceil \frac{L}{2^j} \right\rceil \leq a \leq \left\lceil \frac{L}{2^{j-1}} \right\rceil - 1.$$

This interval contains exactly  $c_j$  integers. These intervals partition  $\{1, \dots, L - 1\}$ .

Reflection  $a \mapsto e - a$  is a bijection from the lower transient exponents to  $\{U + 1, \dots, e - 1\}$ . By (6), it increases every depth by one. The recurrent set contributes  $R$  states of depth zero, and the remaining

exponent  $e$  contributes one state of depth one. Therefore each lower layer contributes  $c_j z^j$ , its reflected layer contributes  $c_j z^{j+1}$ , and summing all contributions gives

$$D_e(z) = R + z + \sum_{j=1}^m c_j (z^j + z^{j+1}),$$

which is (9). □

For example, the exact sharp profile at  $e = 128$  is

$$D_{128}(z) = 44 + 22z + 32z^2 + 16z^3 + 8z^4 + 4z^5 + 2z^6 + z^7,$$

and exponent 127 is the unique depth-seven state.

## 5. IMAGE AND EVERY-TARGET FIBRES

**Theorem 5.1** (Two inverse branches). *Equations (10) and (11) hold for every target  $0 \leq b \leq e$ .*

*Proof.* The doubling branch  $T_e(a) = 2a$  applies precisely when  $3a \leq e$ . Consequently it reaches a target  $b$  precisely when  $b$  is even,  $a = b/2$ , and

$$3b \leq 2e.$$

The reflection branch  $T_e(a) = e - a$  applies precisely when  $3a \geq e$ . Its only possible source over  $b$  is  $a = e - b$ , and the branch condition is again

$$3(e - b) \geq e \iff 3b \leq 2e.$$

For integer  $b$ , this common inequality is equivalent to  $b \leq \lfloor 2e/3 \rfloor = U$ . Hence every  $b \leq U$  has the reflection source  $e - b$  and, when  $b$  is even, the doubling source  $b/2$ ; no  $b > U$  has a source. This proves (11). Since the reflection source exists for every  $0 \leq b \leq U$ , it also proves (10).

Finally, the two candidate sources agree exactly when  $e - b = b/2$ , equivalently  $3b = 2e$ . Treating (11) as a set union therefore gives the correct singleton in the coincident-branch case. □

Transporting (11) back through  $p^a \leftrightarrow a$  gives the literal divisor fibre

$$F_{p,e}^{-1}(p^b) = \{p^{e-b}\} \cup (\{p^{b/2}\} \text{ if } 2 \mid b) \quad (0 \leq b \leq U),$$

and the empty set for  $b > U$ .

## 6. EXACT CONTROLS, ZERO-CREDIT BOUNDARY, AND HOLD

The local verifier recomputes the integer gcd in (1) for every state with  $p \in \{3, 5, 7, 11\}$  and  $2 \leq e \leq 128$ . It then builds each complete functional graph and checks recurrence, fixed iterates through  $k = 12$ , every pointwise time, the temporal polynomial, the image, and every target fibre. A separate binary lane checks every state for  $2 \leq e \leq 48$ , including all equal-valuation exceptions. The audit uses exact integers without sampling, floating point, a computer algebra system, or a third-party package.

| control                                     | exact count |
|---------------------------------------------|-------------|
| odd-prime parameter boxes                   | 508         |
| odd-prime states, orbits, and target fibres | 33,528 each |
| binary parameter boxes / states             | 47 / 1,222  |
| binary equal-valuation cases                | 16          |
| Boolean assertions                          | 319,074     |

TABLE 1. Dependency-free falsification controls. These counts are not proofs of the all-parameter statements and are not ownership evidence.

The conceptual boundary is equally important. General valuation rules, ceiling-log algebra, functional-graph and zeta identities, and formal generating functions are standard tools and receive zero credit. General piecewise-monotone theory [3], reversible discrete tent-map cycle studies [2], and bijective finite skew-tent constructions [1] also receive zero credit. The normalized real map  $x \mapsto \min(2x, 1 - x)$  already exposes the doubling/reflection silhouette.

The residual is therefore narrow: the literal odd-prime divisor-gcd system together with its complete arithmetic temporal and inverse atlas. A bounded search did not locate a direct printed owner for that exact conjunction, but absence from a bounded search is not novelty, priority, authorship, or ownership evidence. If a direct owner is found, or if specialist review judges the arithmetic carrier to be merely decorative, the residual should be killed rather than broadened. No external posting, contact, submission, or release is authorized; status remains `HOLD_EXTERNAL`.

## REFERENCES

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